Growing up in India, where English is often regarded as the language of opportunity, I witnessed first-hand how linguistic hierarchies can quietly shape access to education and technology. This divide exists within Indian languages too. For instance, Hindi's dominance has marginalized its dialects like Bhojpuri, reinforcing inequities by excluding speakers whose languages lack digital or institutional support. Many such underrepresented languages are oral-first. Ironically, **while speech is the most natural form of communication for humans, it remains the modality where machines struggle the most.** This gap motivates the central question driving my research: **How can we build technologies that meaningfully support linguistic diversity, rather than constrain it to the patterns of high-resource languages?** My research therefore, aims to develop speech and language technologies that are inclusive, accessible, and socially responsible, bringing machine understanding closer to the richness of human linguistic and cultural diversity.

To move closer to this overarching goal, during my PhD I wish to:

1. Design **multimodal benchmark** and **evaluation frameworks** that can **measure fairness, representational balance, and cultural understanding**.
2. Democratize **speech** and **text data for low-resource languages** by developing scalable, synthetic data frameworks.
3. Address the fundamental **knowledge misalignment between speech and text modalities** in multimodal LLMs.

My career trajectory has been shaped by the desire to build technology that uplifts people. During my Bachelor's, advised by Prof. Pravesh Biyani, I worked on enhancing routing algorithms for public transportation which is now deployed across various public transportation applications in India. This experience sparked my interest in building efficient, real-world systems and motivated me to explore software engineering. At Microsoft, I strengthened my technical foundations by developing robust, production-grade systems. As language technologies gained momentum, their possibilities for Indian languages in sectors like education and agriculture fascinated me. Therefore, I joined AI4Bharat, a research lab at IIT Madras, where I'm now pursuing a research-focused Master's degree under the guidance of Prof. Mitesh Khapra. Through my projects and foundational coursework at IITM, I have developed deep appreciation for research and I believe a PhD is the natural next step in my journey. Below, I outline my work and future interests.

Enhancing Low-Resource Speech Systems

Identifying the weaknesses of a system is the first step toward making it better. Working with Indian Text-to-Speech (TTS) systems, I observed that they consistently mispronounced out-of-domain and code-mixed words, common in everyday use. This gap stemmed less from modeling choices and more from data sparsity, especially missing character bigrams in existing corpora. To study this systematically, I led the project **IndicOOV**[1], where I showed that targeted volunteer-recorded speech addressing these gaps could significantly improve intelligibility. Driving this project from early problem definition to solution development helped shape my approach to research and strengthened my skills in experiment design, dataset development, and comprehensive evaluation using both human and automated methods. IndicOOV was accepted for oral presentation at **INTERSPEECH 2024** and was selected for presentation at the **Young Female Researchers in Speech Workshop 2024**.

With IndicOOV's success, I became interested in exploring scalable data-centric methods to improve Indian TTS systems. Two questions followed, first, how can we use the more abundant noisy Automatic Speech Recognition(ASR) data for TTS and second, how different data augmentation strategies influence intelligibility of TTS systems and whether their impact can be measured systematically. In **IndicVoices-R**[2]**(NeurIPS 2024, Datasets and Benchmark Track)**, we developed a speech-restoration pipeline, restoring more than 1700 hours of speech to high-quality TTS data, across 22 Indian languages. The work was very special as we got the first TTS systems for eight Indian languages. We proposed two metrics in **ELAICHI**[3] (**submitted to ICASSP 2026**), that help quantify why certain data interventions work and improve intelligibility more effectively than others. In ELAICHI, I witnessed the astonishing power of synthetic data as with just 25 hours of speech we achieved comparable intelligibility to our best performing setup with over 600 hours of training data.

While building TTS systems, I realized that meaningful progress in the field depends as much on eval-

uation as on modeling. Though TTS depends largely on human evaluations, widely used frameworks such as MOS, CMOS, and even MUSHRA are coarse and opaque, offering limited insight into the multi-dimensional nature of speech quality. This gap motivated **Rethinking MUSHRA**[4](**TMLR 2025**), where we designed finer-grained evaluation strategies that reduce reference-matching bias and capture more nuanced listener preferences. Collectively, these projects gave me a holistic and deep understanding of speech synthesis. However, I also began thinking beyond improving speech outputs and toward a broader question: **how do models see, interpret, and reason about our world?**

Aligning Multimodal Intelligence With Global Cultures

The shift in my research interests and a growing desire to explore research beyond the Indian languages led to my discussions with Prof. Thamar Solorio whose work on Cultural Visual Question Answering[5] deeply resonated with me. This culminated in a **Summer 2025 research internship at MBZUAI**, where we developed **Afri-MCQA** (**submitted to EACL 2026**), an audio-visual cultural question answering benchmark spanning 15 African languages. The project evaluated omni-LLMs in a challenging multimodal (speech + text) and multilingual setup, testing their ability to answer culturally grounded questions about images. I led the technical development of the benchmark, including designing the data pipeline, experimenting with omni-LLMs, and developing the evaluation strategies. Working closely with annotators gave me valuable exposure to linguistic structures, cultural norms, and communicative practices beyond the Indian context, insights that shaped decisions in dataset design and evaluation protocol and helped us better understand the evaluation outcomes.

This experience crystallized two key problems. First, **evaluation**: while benchmarks often rely on Multiple Choice Question Answering (MCQA) to evaluate LLM knowledge, it is prone to selection and position bias [6] and does not give a clear picture of whether the model truly understands the question, context and the answer options making it difficult to draw insights about the LLM's knowledge[7]. We explored answer generation and LLM-as-a-judge setups, but found that the LLMs' limited ability to comprehend and generate text in low-resource languages undermines the reliability of such evaluations, making them a weak indicator of the model's true knowledge.

Second, **alignment failure**: The evaluations provided stark evidence of a two-fold misalignment - a cross-lingual gap between native languages and English and a cross-modal gap between speech and text modalities. While this has been documented[8], alignment efforts have focused on high-resource languages[9], they often presume the availability of large-scale parallel corpora, which is not feasible for oral-first languages. The misalignment therefore, is especially critical for these communities, where the modality gap effectively locks vast repositories of cultural knowledge away from text-centric models, hampering downstream tasks and practical applications. Inspired by architectures like SpiritLM[9], it will be interesting to investigate whether interleaving high-resource text tokens with low-resource speech tokens while training can encourage alignment in the shared embedding space, enabling reasoning in oral-first languages without relying on large parallel corpora.

Why University of Maryland?

I am particularly excited to work under the guidance of **Prof. Marine Carpuat** as our interests align closely on human-centered and multilingual AI. I am also genuinely interested in working with **Prof. Mohit Iyyer** and **Prof. Jordan Boyd-Graber**. Prof. Iyyer's work CaLMQA[10] was influential during my work on Afri-MCQA and helped shape my understanding of cultural question-answering. I found Prof. Graber's works on consistent multilingual question-answering[11] and addressing flaws of MCQ evaluations for LLMs[12] in line with my interests in enhancing evaluation frameworks for LLMs. Beyond PhD, I am interested in continuing research work on socially inclusive and responsible AI. I am equally interested in working under the guidance of **Prof. Ramani Duraiswami** as his expertise on speech and audio language models would very valuable in my study of multimodal benchmarking and reducing alignment gaps in speech and text. Beyond PhD, I want to work on research problems at the intersection of AI and society, while also building products that translate this research into real-world impact. Therefore, the collaborative and interdisciplinary community at University of Maryland, with its emphasis on human-centered AI, makes it the perfect place for me to pursue my PhD. Based on my research experiences and interests, I am confident that I can positively contribute to the vibrant NLP and speech community at the university.

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